SBU AIC Datathon

Interest, Inflation, & Irritation

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Introduction

01. Context

Macroeconomic conditions such as interest rates and inflation shape everyday financial life in the United States. They influence borrowing costs, repayment pressure, and household decision-making across products such as credit cards, mortgages, student loans, and auto loans.

02. The Research Question

How do changes in inflation and interest rates relate to complaint activity across major consumer financial products?

03. Our Hypothesis

Credit cards are most sensitive to economic stress, especially interest rate spikes.

The Datasets

Interest Rate

FEDFUNDS (U.S. Interest Rate Dataset)

This dataset reports the Effective Federal Funds Rate, which is the overnight interest rate at which depository institutions lend reserve balances to one another. It is a widely used monthly indicator of U.S. monetary policy and overall interest-rate conditions.

Inflation Rate

CPIAUCSL (U.S. Inflation Dataset)

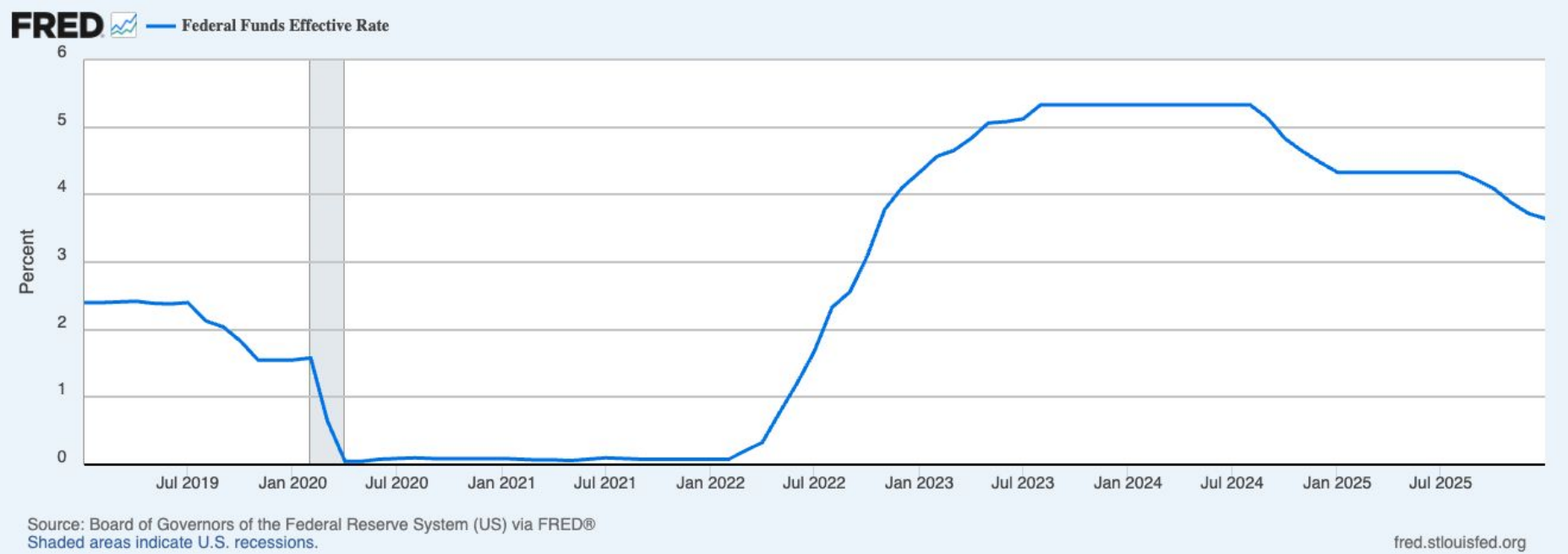
This dataset measures the Consumer Price Index for All Urban Consumers and tracks changes in the prices paid by urban consumers for a basket of goods and services. It is a standard monthly measure of U.S. inflation, and percent changes in this index are commonly used to calculate inflation rates over time.

Consumer Complaints

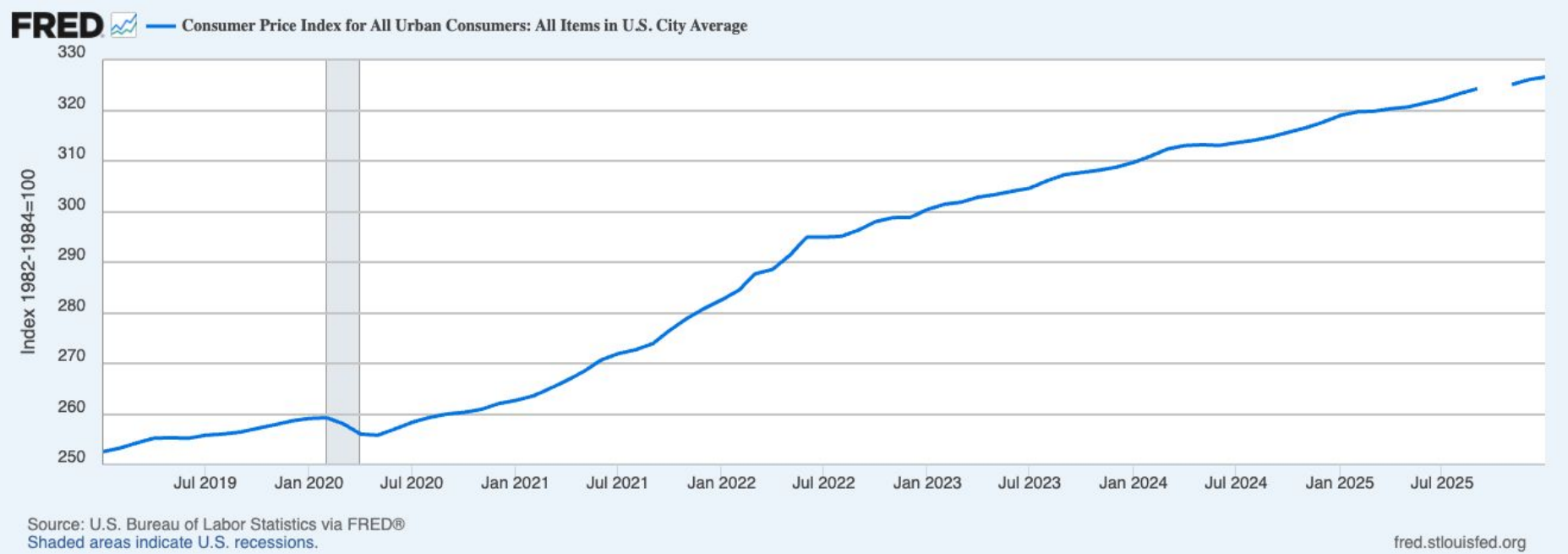
CFPB Consumer Complaint Database

This dataset contains public complaints submitted about consumer financial products and services, including products such as credit cards, mortgages, checking accounts, student loans, and vehicle loans. It includes fields such as date received, product, issue, company, and company response, making it useful for analyzing how consumer financial stress changes over time.

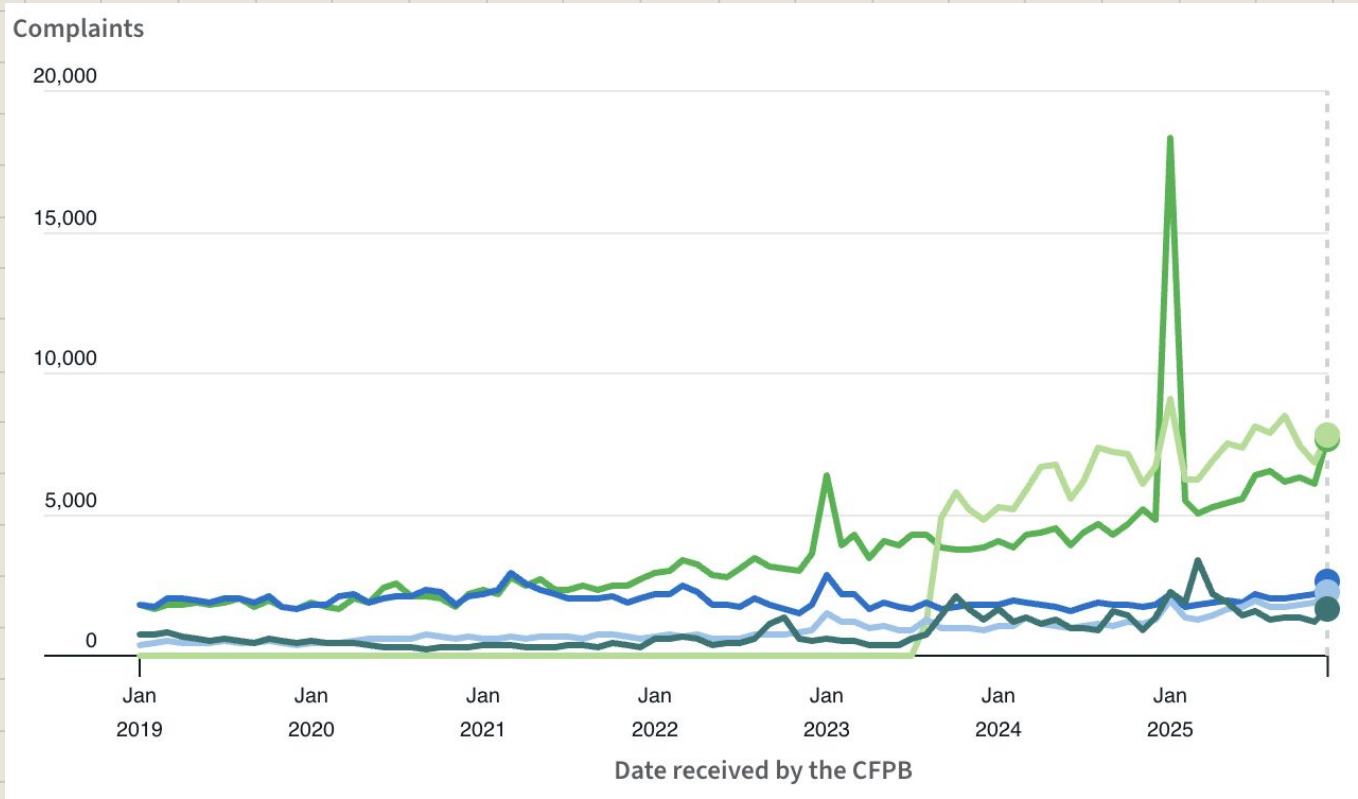
Interest Rates



Inflation

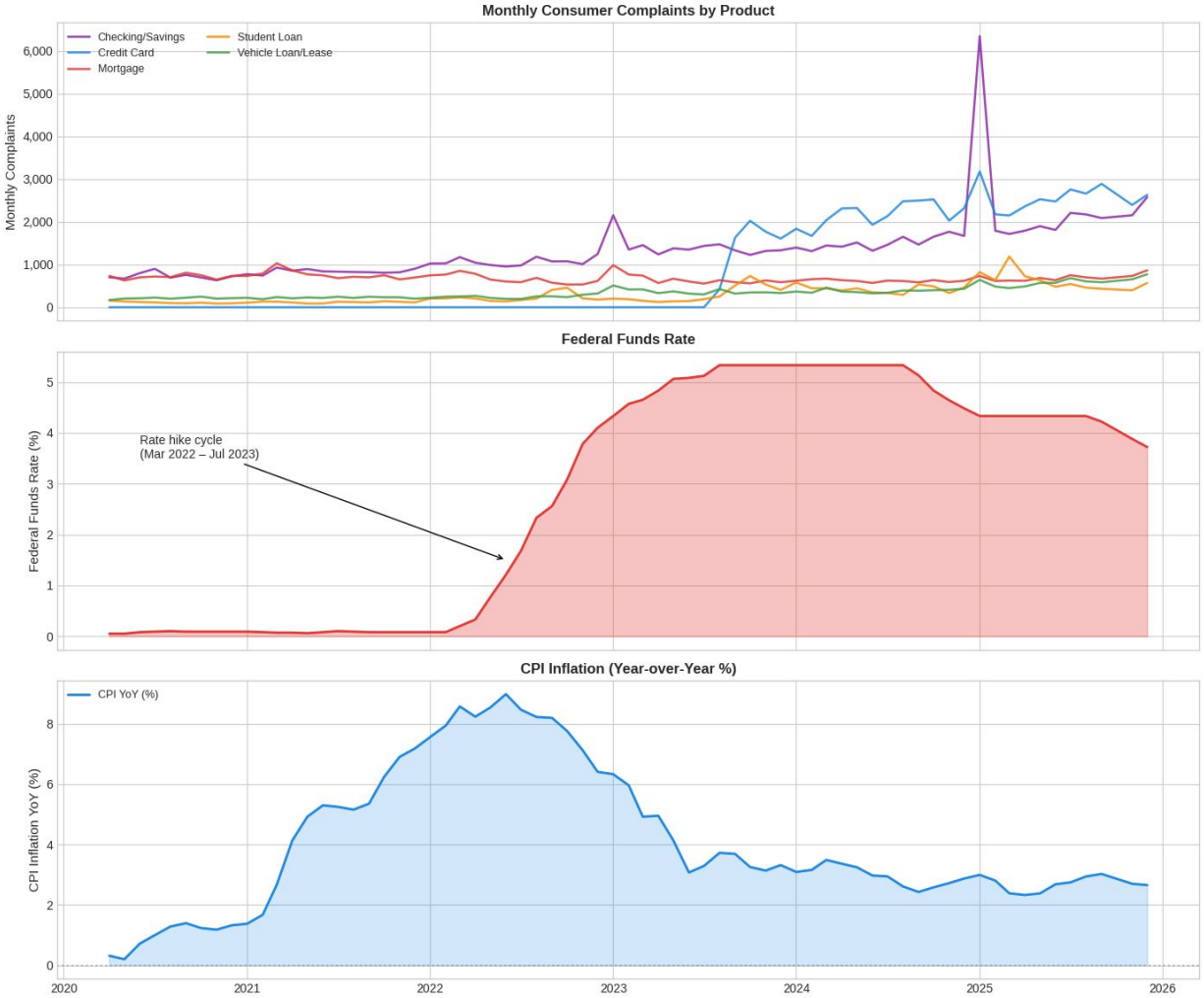

FRED
Consumer Price Index for All Urban Consumers: All Items in U.S. City Average
Index 1982-1984=100
330
320
310
300
290
280
270
260
250
Jul 2019
Jan 2020
Jul 2020
Jan 2021
Jul 2021
Jan 2022
Jul 2022
Jan 2023
Jul 2023
Jan 2024
Jul 2024
Jan 2025
Jul 2025
Source: U.S. Bureau of Labor Statistics via FRED®
Shaded areas indicate U.S. recessions.
fred.stlouisfed.org

Consumer Complaints



- Checking or savings account
- Credit card
- Mortgage
- Vehicle loan or lease
- Student loan

Consumer Complaints & Economic Stress (2019–2025)

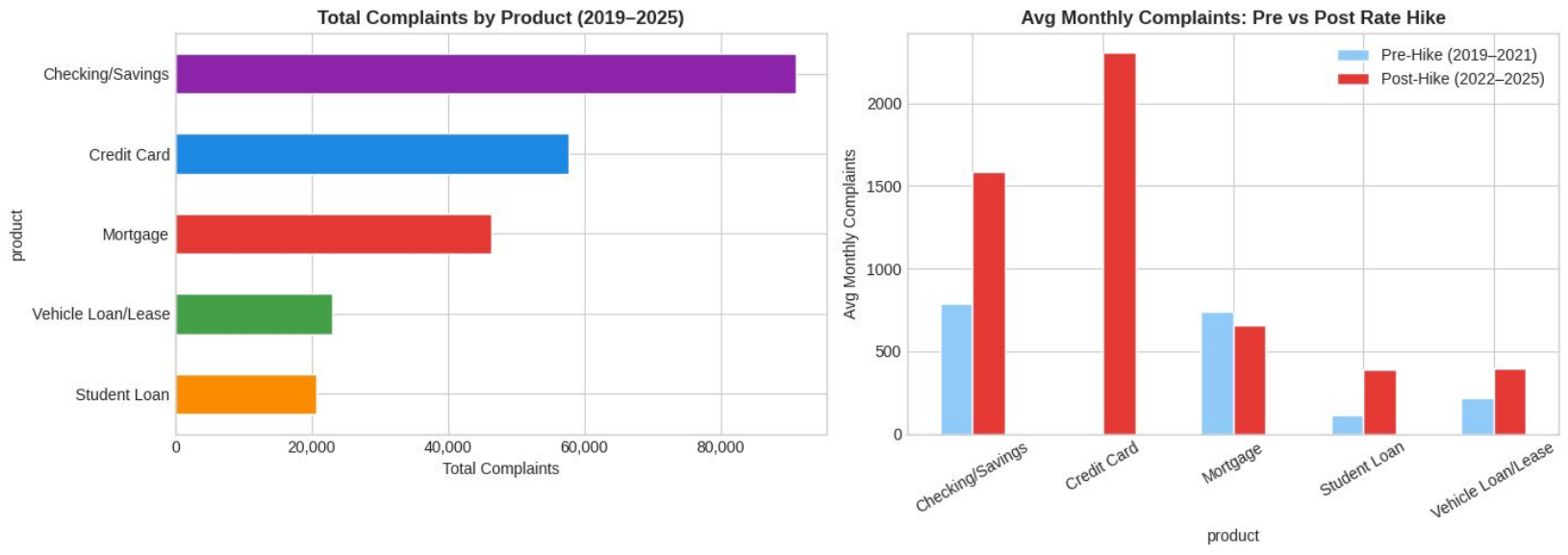


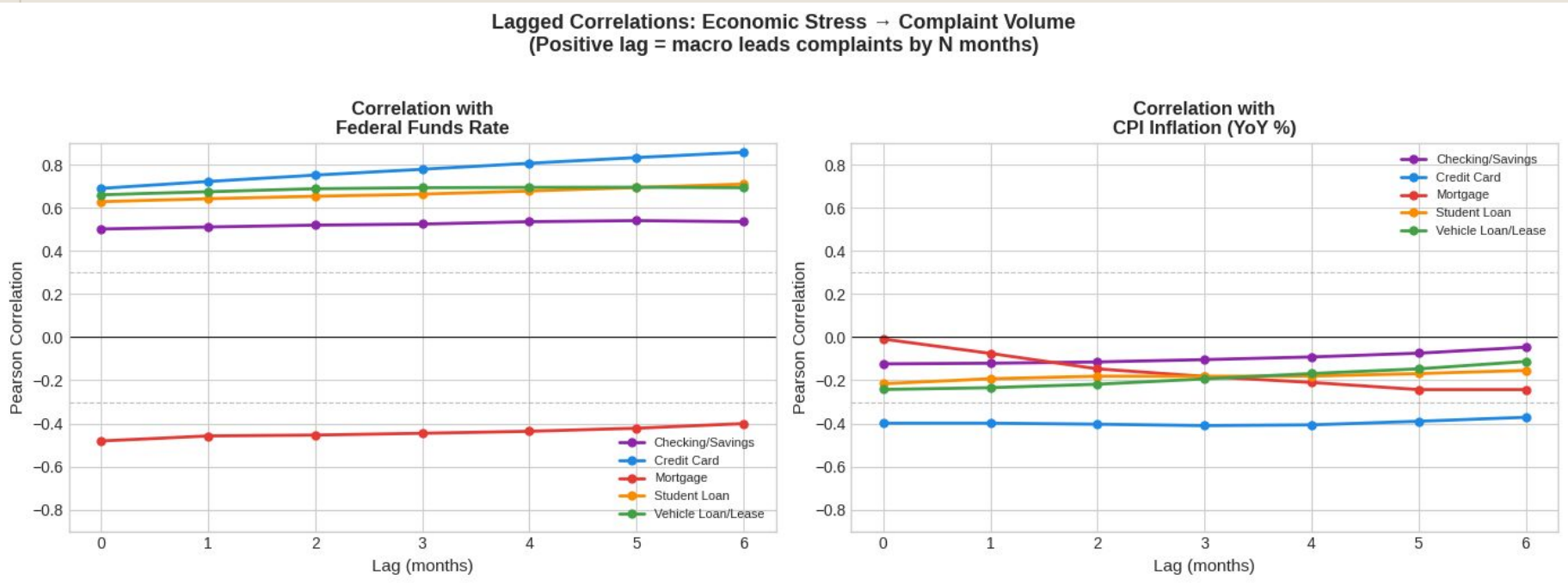
Data Visualization Code

```python
PRODUCT_COLORS = {
    'Mortgage': '#E53935',
    'Credit Card': '#1E88E5',
    'Student Loan': '#FB8C00',
    'Checking/Savings': '#8E24AA',
    'Vehicle Loan/Lease': '#43A047'
}

fig, axes = plt.subplots(3, 1, figsize=(14, 12), sharex=True)

plot1 = axes[0]
for product in products:
    subset = panel_long[panel_long['product'] == product]
    color = PRODUCT_COLORS.get(product, 'gray')
    plot1.plot(subset['month'], subset['complaint_count'], label=product, color=color, linewidth=1.8, alpha=0.85)
plot1.set_ylabel('Monthly Complaints', fontsize=11)
plot1.set_title('Monthly Consumer Complaints by Product', fontsize=12, fontweight='bold')
plot1.legend(loc='upper left', fontsize=9, ncol=2)
plot1.yaxis.set_major_formatter(mticker.FuncFormatter(lambda x, _: f'{x:,.0f}'))


plot2 = axes[1]
rate_data = panel_long[panel_long['product'] == products[0]].sort_values('month')
plot2.fill_between(rate_data['month'], rate_data['fedfunds_rate'], alpha=0.3, color='#E53935')
plot2.plot(rate_data['month'], rate_data['fedfunds_rate'], color='#E53935', linewidth=2)
plot2.set_ylabel('Federal Funds Rate (%)', fontsize=11)
plot2.set_title('Federal Funds Rate', fontsize=12, fontweight='bold')
plot2.annotate('Rate hike cycle\n(Mar 2022 – Jul 2023)', xy=(pd.Timestamp('2022-06-01'), 1.5), xytext=(pd.Timestamp('2020-06-01'), 3.5),
               arrowprops=dict(arrowstyle='->', color='black'))


plot3 = axes[2]
plot3.fill_between(rate_data['month'], rate_data['inflation_yoy'], alpha=0.2, color='#1E88E5')
plot3.plot(rate_data['month'], rate_data['inflation_yoy'], color='#1E88E5', linewidth=2, label='CPI YoY (%)')
plot3.set_ylabel('CPI Inflation YoY (%)', fontsize=11, color='#1E88E5')
plot3.set_title('CPI Inflation (Year-over-Year %)', fontsize=12, fontweight='bold')
plot3.axhline(0, color='gray', linestyle=':', alpha=0.5)
plot3.legend(loc='upper left', fontsize=9)

plt.suptitle('Consumer Complaints & Economic Stress (2019–2025)', fontsize=15, fontweight='bold', y=1.01)
plt.tight_layout()
plt.show()
```

Findings

```python
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

total_by_product = (panel_long.groupby('product')['complaint_count'].sum().sort_values(ascending=True))
colors = [PRODUCT_COLORS.get(p, 'gray') for p in total_by_product.index]
total_by_product.plot(kind='barh', ax=axes[0], color=colors, edgecolor='white')
axes[0].set_title('Total Complaints by Product (2019–2025)', fontsize=12, fontweight='bold')
axes[0].set_xlabel('Total Complaints')
axes[0].xaxis.set_major_formatter(mticker.FuncFormatter(lambda x, _: f'{x:,.0f}'))

pre = panel_long[panel_long['year'] <= 2021].groupby('product')['complaint_count'].mean()
post = panel_long[panel_long['year'] >= 2022].groupby('product')['complaint_count'].mean()
comparison = pd.DataFrame({'Pre-Hike (2019–2021)': pre, 'Post-Hike (2022–2025)': post})
comparison.plot(kind='bar', ax=axes[1], color=['#90CAF9', '#E53935'], edgecolor='white')
axes[1].set_title('Avg Monthly Complaints: Pre vs Post Rate Hike', fontsize=12, fontweight='bold')
axes[1].set_ylabel('Avg Monthly Complaints')
axes[1].tick_params(axis='x', rotation=30)
axes[1].legend()

plt.tight_layout()
plt.show()

print('\nPercentage change pre vs post rate hike:')
percentage_change = ((post - pre) / pre * 100).replace([np.inf, -np.inf], np.nan).dropna().sort_values(ascending=False)
for product, value in percentage_change.items():
    print(f'  {product:25s}: {value:+.1f}%')
```

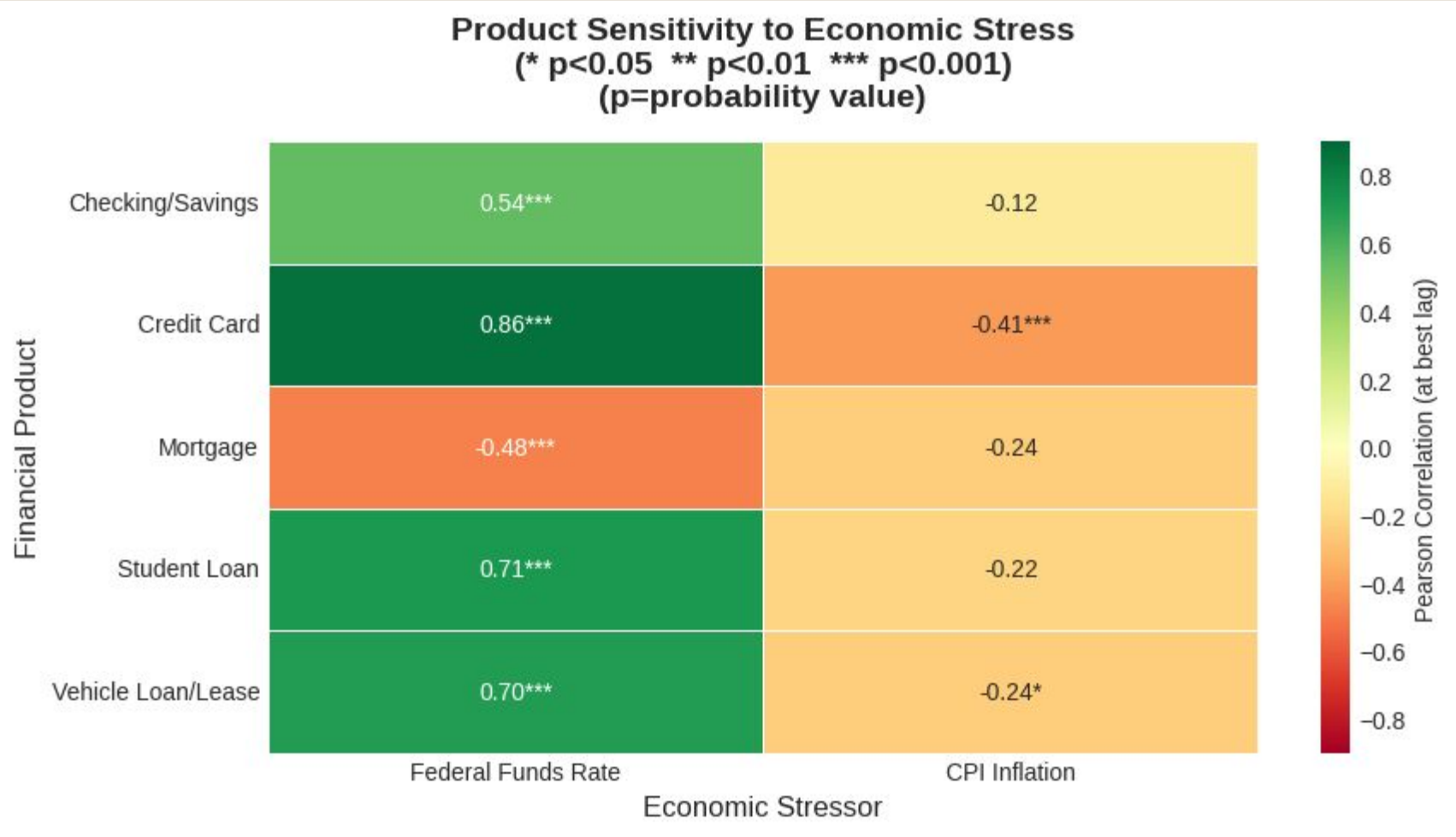



Lagged Correlations: Economic Stress → Complaint Volume
(Positive lag = macro leads complaints by N months)
Correlation with
Federal Funds Rate
Pearson Correlation
Lag (months)
Checking/Savings
Credit Card
Mortgage
Student Loan
Vehicle Loan/Lease
Correlation with
CPI Inflation (YoY %)
Pearson Correlation
Lag (months)
Checking/Savings
Credit Card
Mortgage
Student Loan
Vehicle Loan/Lease

Lagged Correlation Code

```python
MACRO_VARS = ['fedfunds_rate', 'inflation_yoy']

def compute_lagged_correlations(panel_long, products, macro_vars, max_lag=6):
    results = []
    for product in products:
        subset = panel_long[panel_long['product'] == product].sort_values('month')
        for macro in macro_vars:
            if macro not in subset.columns:
                continue
            for lag in range(0, max_lag + 1):
                complaint_s = subset['complaint_count'].values
                macro_s = subset[macro].shift(lag).values
                mask = ~np.isnan(macro_s) & ~np.isnan(complaint_s)
                if mask.sum() < 10:
                    continue
                correlation, p_vlaue = stats.pearsonr(complaint_s[mask], macro_s[mask])
                results.append({
                    'product': product, 'macro_var': macro, 'lag': lag, 'correlation': correlation,
                    'p_value': p_value, 'significant': p_value < 0.05
                })
    return pd.DataFrame(results)


lag_results = compute_lagged_correlations(panel_long, products, MACRO_VARS, max_lag=6)

print(f'Lag correlations computed: {len(lag_results)} combinations')

best_lags = (
    lag_results
    .assign(abs_corr=lambda x: x['correlation'].abs())
    .sort_values('abs_corr', ascending=False)
    .groupby(['product', 'macro_var'], as_index=False)
    .first()
    .drop(columns='abs_corr')
    .sort_values(['macro_var', 'correlation'], ascending=[True, False])
)

print('\nBest lag for each product × stressor:')
print(best_lags[['product','macro_var','lag','correlation','p_value']].to_string(index=False))
```

```python
macro_titles = {'fedfunds_rate': 'Federal Funds Rate', 'inflation_yoy': 'CPI Inflation (YoY %)'}

fig, axes = plt.subplots(1, 2, figsize=(14, 5), sharey=False)

for ax, macro in zip(axes, MACRO_VARS):
    subset = lag_results[lag_results['macro_var'] == macro]
    for prod in products:
        prod_data = subset[subset['product'] == prod].sort_values('lag')
        if prod_data.empty:
            continue
        color = PRODUCT_COLORS.get(prod, 'gray')
        ax.plot(prod_data['lag'], prod_data['correlation'],
                marker='o', label=prod, color=color, linewidth=2, markersize=5)
    ax.axhline(0,    color='black', linewidth=0.8)
    ax.axhline(0.3,  color='gray',  linewidth=0.8, linestyle='--', alpha=0.5)
    ax.axhline(-0.3, color='gray',  linewidth=0.8, linestyle='--', alpha=0.5)
    ax.set_xlabel('Lag (months)', fontsize=11)
    ax.set_ylabel('Pearson Correlation', fontsize=11)
    ax.set_title(f'Correlation with\n{macro_titles[macro]}', fontsize=12, fontweight='bold')
    ax.set_xticks(range(0, 7))
    ax.legend(fontsize=8)
    ax.set_ylim(-0.9, 0.9)

plt.suptitle('Lagged Correlations: Economic Stress → Complaint Volume\n(Positive lag = macro leads complaints by N months)',
             fontsize=13, fontweight='bold', y=1.02)
plt.tight_layout()
plt.show()
```


Product Sensitivity to Economic Stress
(* p<0.05 ** p<0.01 *** p<0.001)
(p=probability value)
Financial Product
Checking/Savings
Credit Card
Mortgage
Student Loan
Vehicle Loan/Lease
0.54***
-0.12
0.86***
-0.41***
-0.48***
-0.24
0.71***
-0.22
0.70***
-0.24*
Federal Funds Rate
CPI Inflation
Economic Stressor
Pearson Correlation (at best lag)
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8

```python
sensitivity_matrix = best_lags.pivot(index='product', columns='macro_var', values='correlation')

col_rename = {'fedfunds_rate': 'Federal Funds Rate', 'inflation_yoy': 'CPI Inflation'}
sensitivity_matrix.columns = [col_rename.get(c, c) for c in sensitivity_matrix.columns]

pval_matrix = best_lags.pivot(index='product', columns='macro_var', values='p_value')

annot_matrix = sensitivity_matrix.copy().astype(str)
for product in sensitivity_matrix.index:
    for disp_col, raw_col in [('Federal Funds Rate','fedfunds_rate'), ('CPI Inflation','inflation_yoy')]:
        if disp_col in sensitivity_matrix.columns and raw_col in pval_matrix.columns:
            correlation = sensitivity_matrix.loc[product, disp_col]
            p_value = pval_matrix.loc[product, raw_col]
            stars = '***' if p_value < 0.001 else '**' if p_value < 0.01 else '*' if p_value < 0.05 else ''
            annot_matrix.loc[product, disp_col] = f'{correlation:.2f}{stars}'

fig, ax = plt.subplots(figsize=(9, 5))
sns.heatmap(sensitivity_matrix,vannot=annot_matrix, fmt='',vcmap='RdYlGn', center=0, vmin=-0.9, vmax=0.9,
    linewidths=0.5, linecolor='white', ax=ax, cbar_kws={'label': 'Pearson Correlation (at best lag)'})
ax.set_title('Product Sensitivity to Economic Stress\n(* p<0.05  ** p<0.01  *** p<0.001)\n(p=probability value)',
             fontsize=14, fontweight='bold', pad=15)
ax.set_xlabel('Economic Stressor', fontsize=12)
ax.set_ylabel('Financial Product', fontsize=12)
ax.tick_params(axis='x', rotation=0)
ax.tick_params(axis='y', rotation=0)
plt.tight_layout()
plt.show()
print('\nSensitivity Matrix:')
print(sensitivity_matrix.round(3))
```

Key Takeaways

1. **Interest rates matter more than inflation in our analysis**
    a. Credit Card has a very strong positive association with the Fed Funds Rate
    b. Student Loan and Vehicle Loan/Lease are also strongly positive
    c. Checking/Savings is moderately positive
2. **Different products respond differently to macro stress**
    a. Student Loan and Vehicle Loan/Lease appear highly rate-sensitive
    b. Checking/Savings is positive but less extreme
    c. Mortgage behaves differently from the other debt products
    d. Credit Card is especially sensitive, but the sign depends on the modeling setup and needs careful explanation
3. **There is evidence of lagged effects**
    a. Our lag analysis suggests that some products respond several months after the macro stressor changes, not immediately.

Limitations

- **Correlation does not mean causation**
  - Our findings identified associations between macroeconomic indicators (CPI inflation, Fed interest rate) and consumer complaint volumes, but they cannot prove causation.
- **Complaint counts are not the same as actual consumer distress**
  - A consumer can file a complaint only if they have knows where and how to file it.
  - Complaint volume may reflect consumer awareness and reporting behavior.
- **Raw complaint counts are not normalized by market size**
  - Because we use raw complaint counts rather than complaint rates, some results may reflect changes in market activity rather than changes in consumer stress per user.
- **Limited macro variables**
  - We only used interest rate and inflation data.
- **Time period contains unusual events**
  - We customized our datasets from Jan 2019 to Dec 2025 which included the pandemic, rapid inflation, and aggressive rate hikes.

Datasets Sources

- https://www.consumerfinance.gov/data-research/consumer-complaints/
- https://fred.stlouisfed.org/series/CPIAUCSL
- https://fred.stlouisfed.org/series/FEDFUNDS

Thank You!